Symmetrix-Polar

Consolidated speed and accuracy comparison

MACE-Polar (PolarMACE) on the Symmetrix-XL nonlinear Kokkos engine

LAMMPS pair style, single A100, f64

Version: 2026-10-19

Model: polar_symmetrix_probe.json (MACE_Nonlinear + polar)

Repositories: Yong-Q/symmetrix-polar · starlessinred-interns/qiuyong

This document is generated from docs/COMPARISON_REPORT_2026-10-19.md. All numbers come from the repository JSON/log artifacts; in-flight results are marked pending. The force gate (1e-4 eV/Å) is treated as a hard minimum: RBE and 1-hop ghost MPI are rejected as production alternatives.

Symmetrix-Polar: Consolidated Speed and Accuracy Comparison

Date: 2026-10-19 (v29 / v30 evidence cut-off, ~08:20 UTC) **Scope:** PolarMACE (MACE-Polar) support in the Symmetrix-XL C++/Kokkos engine. **Basis:** completed JSONs, gate logs, and reports only. No GPU benchmark was run for this document, and no engine/model code was modified.

Status legend: PASS = measured and accepted against its gate; **FAIL** = measured and rejected; **pending** = work is in flight and no completed artifact exists yet. No pending result is included as a performance claim.

0. Summary

Item	Result
Single-rank v29 speed, 128 atoms, deterministic	183.78 ms/eval (v28: 6966.79 ms, 37.91x)
Single-rank v29 speed, 1024 atoms, deterministic	1007.72 ms/eval (v28: 7795.59 ms, 7.74x)
Same-model ablation, 128 atoms	backbone-only 87.24 ms -> full PolarMACE 183.50 ms; polar layer adds 96.26 ms (2.10x)
Accuracy gates, modules 5-8	all PASS
End-to-end engine vs PyTorch, 32 frames	32/32 PASS, max energy error 1.034e-6 eV, max force error 1.624e-7 eV/A
LAMMPS single point, 1 rank / exact 2 rank	PASS / PASS (dE 2.91e-11 eV; max dF 2.62e-14 / 1.69e-6 eV/A)
LAMMPS NVT MD, 50 steps	PASS, no lost atoms, Etot span 0.21295 eV
RBE production use	rejected by the force gate (RMS_F 4.10e-2 to 1.03e-2 eV/A vs 1e-4 eV/A gate)
1-hop MPI mode	rejected by the force/energy gates (max dF 4.116e-3 eV/A, dE 7.94e-5 eV)
MPI speed on one shared A100	no speedup; exact 2-hop is 28.1x slower than 1 rank
MH-1/direct/JIT fast path	architecturally blocked; the upstream linear engine cannot load MACE_Nonlinear
PyTorch MACE-Polar reference speed	pending (script added; no log yet)

The v29 optimization removed two concrete bottlenecks: per-node-serial UVU tensor products and a ~0.7 s/eval device-environment probe on non-JIT models. All numerical gates are unchanged after the optimization.

1. Scope and model

1.1 What is being compared

All "same-model" statements below compare the same PolarMACE checkpoint (`MACE-POLAR-1-M.model`) exported to the engine JSON `polar_symmetrix_probe.json`, evaluated in the same 128-atom NH3 box (14.736126 A cubic, periodic) with the same float64 energy+force path. The linear MACE-OFF23 comparison is explicitly **not** the same model and is labelled wherever it appears.

Property	Full PolarMACE	Nonlinear backbone (polar off)	Linear MACE-OFF23 medium
JSON model_type	MACE_Nonlinear	MACE_Nonlinear	MACE
Engine	MaceNonlinearKokkosT	MaceNonlinearKokkosT	MACEKokkos
JSON size (bytes)	316,958,257	269,988,335	37,330,647
Elements	83	83	10
r_cut	6.0 A	6.0 A	5.0 A
Interactions / l_max	2 / 3	2 / 3	standard MACE
Channel pattern	512 -> 128, nonlinear bias readout	512 -> 128, nonlinear bias readout	linear readout
Polar layer	full SCF + Ewald/RBE	disabled	inert placeholder

Property	Full PolarMACE	Nonlinear backbone (polar off)	Linear MACE-OFF23 medium
Execution mode	materialized (no direct/JIT)	materialized	materialized or generic streamed
Typical use here	production target	same-model backbone ablation	engine-path reference only

Hardware/software: one NVIDIA A100-80GB per run, CUDA 12 sm80 backend, Kokkos 5.x, float64, energy+forces. Times are LAMMPS `Loop time` or Python `median_ms` over 20 evals, as noted. The GPU was shared during part of the campaign; the v29 final figures are the uncontended reruns. Absolute times are therefore contention-sensitive, while same-run ratios are stable.

1.2 Linear vs nonlinear engines

"Linear" and "nonlinear" here are Symmetrix engine/model families, not linear-vs-nonlinear MLIPs:

	MACEKokkos (linear engine)	MaceNonlinearKokkosT (nonlinear engine)
Accepted model_type	MACE, MACEField	MACE_Nonlinear
JSON format	format-version 2 compact radial	format-version 3 nonlinear
Readout	standard linear MACE readout	nonlinear readout (nonlinear, NonLinearBiasReadoutBlock for PolarMACE)
Polar blocks	polar_blocks_enabled() returns false	full polar block set and SCF pipeline
Direct gate	supports_streamed_edges() = uses_compact_radial	supports_streamed_edges() = mh1_fast_path
H1 MPI	first-interaction hidden node state	materialized node-feature exchange only

The upstream linear Kokkos engine cannot load MACE_Nonlinear at all. `MACEKokkos::load_from_json` rejects it at `mace_kokkos_model.cpp:165-167` with:

```
MACE_Nonlinear JSON must be loaded through the nonlinear MACE Kokkos evaluator,
not legacy MACEKokkos.
```

The naming "H1" is a collision: in the linear engine H1 is the first-interaction hidden state; it is not the MACE-MH-1 model family. No PolarMACE layer can be delegated to `MACEKokkos`, and the linear engine's `mpi_message_passing` path cannot be reused for PolarMACE.

1.3 MH-1 fast-path status

PolarMACE cannot use the current MACE-MH-1 direct/JIT fast path. The engine's family descriptor rejects it with:

```
MH-1 node, edge, and hidden irreps are inconsistent.
```

This is architectural, not an export defect. The original checkpoint has the same architecture as the JSON. The four deviations from the admitted MH-1 family are:

1. second interaction uses `128x0e+128x1o` edge features instead of the first interaction's `512x0e+512x1o`;
2. second interaction keeps a vector component in `hidden_irreps` (`512x0e+512x1o` instead of scalar-only `512x0e`);
3. second product returns `512x0e+512x1o` instead of scalar-only `512x0e`;
4. second readout is `NonLinearBiasReadoutBlock` (with a learned `linear_mid` stage) instead of `NonLinearReadoutBlock`.

A v3 `MH1_UVU` contract is present in the JSON, but it covers only the per-interaction UVU tensor products and does not establish v4 node-program family compatibility. The v4 contract generator fails first at `products[1] output irreps are outside the MH-1 support matrix`. Re-exporting (Route A) cannot fix this without changing the learned function; relaxing only the gate (Route C) is numerically incorrect and is also not a speed win. See `docs/design/MH1_FASTPATH_FEASIBILITY.md` for the full rejection chain and the staged B1-B4 route assessment. No speedup from MH-1 is claimed anywhere in this report.

2. Accuracy and correctness gates

2.1 Module gates 5-8 and end-to-end

Gate	Metric	Tolerance	Measured	Verdict
Module 5, density (12 items)	max relative density error	rel <= 1e-6	4.22e-8	PASS
Module 5, density (12 items)	max absolute density error	abs <= 5e-6	6.689e-8	PASS
Module 5, stages (11 items)	max relative stage error	rel <= 1e-6	7.13e-8	PASS
Module 5, pairing	engine state -> fixture step mapping	exact	['step1','step2','step2'] for all 4 molecules	PASS
Module 6, Coulomb	max rho-hat relative error	rel <= 1e-6	1.26e-15	PASS
Module 6, Coulomb	max dE relative / absolute	rel <= 1e-6; abs <= 5e-6 eV	8.43e-16 / 5.551e-16 eV	PASS
Module 6, Coulomb	max end-to-end energy error	informational	1.491e-8 eV	PASS
Module 7, local electron	max stage absolute error	abs <= 5e-6 or rel <= 1e-6	2.230e-5 abs (passes on the relative criterion)	PASS
Module 7, local electron	max output absolute error	abs <= 1e-6	4.038e-7	PASS
Module 8, forces	max finite-difference error	<= 1e-3	5.783e-4	PASS
Module 8, forces	max parity vs PyTorch	<= 1e-3	7.191e-7	PASS
Module 8, forces	max sum-of-forces residual	exact translation invariance	1.776e-14	PASS
End-to-end, 32 frames	frames within tolerance	32/32	32/32	PASS
End-to-end, energy	max relative / absolute	rel <= 1e-6; abs <= 5e-6 eV	1.1e-11 / 1.034e-6 eV	PASS
End-to-end, forces	max force error	<= 1e-4 eV/A	1.624e-7 eV/A (mean max 9.456e-8)	PASS
End-to-end, geometry	max position drift	0	0.0 A	PASS

Notes: - Module 5 evaluated 12 density items (4 molecules x 3 gated step comparisons) plus 11 stage tensors. The separate "anchor" field-independent density comparison (rel up to 9.57e-8) is recorded but not part of the module-5 gate. - Module 7 passes because its criterion is `abs <= 5e-6 OR rel <= 1e-6`; the worst absolute stage item (co2-f1, 2.230e-5) is at reference scale 351.146 and passes on the relative criterion. - Module 8 also reports `|polar|max` between 3.0e-3 and 4.7e-2 eV/A across configurations, confirming the polar force term is not negligible.

2.2 RBE statistical gate

Source: `results/polar_rbe_gate_v29.json`, log `polar_rbe_gate_v29opt.log`.

Metric	Tolerance / criterion	Measured	Verdict
Slope of log RMS_F vs log p	ideal -0.5	-0.49885	PASS
Same-seed reproducibility	bitwise energy; dF ~ 0	dE = 0; max dF = 6.141e-16 eV/A	PASS
Different-seed response	non-zero	dE = 0.1003 eV; dF = 9.365e-2	PASS
RMS_F at p = 50	decreasing	4.0988e-2 eV/A	statistical PASS
RMS_F at p = 200	decreasing	2.1426e-2 eV/A	statistical PASS
RMS_F at p = 800	decreasing	1.0280e-2 eV/A	statistical PASS
Mean dE at p = 50 / 200 / 800	unbiased (n=10)	-5.139e-2 / -2.344e-2 / +1.469e-3 eV	PASS
Max abs dE at p = 50 / 200 / 800	monitored	1.890e-1 / 1.426e-1 / 2.936e-2 eV	monitored

The RBE estimator is unbiased and converges at the expected -1/sqrt(p) rate, but its force RMS at every measured p is **two to three orders of magnitude above the 1e-4 eV/A production force gate**. Under the project's gate-as-minimum rule (a speed mode is usable only if every accuracy gate still passes), RBE is accepted as a sampling/convergence gate but **rejected for production force evaluation**. See Section 3.6 for its 33 A timings.

2.3 LAMMPS pair-style and MD gates

Sources: `results/polar_lammps_parity.json`, `results/polar_lammps_parity_mpi2.json`,

`results/polar_md.json`, logs `polar_lammps_parity_v29final.log`, `polar_lammps_md_v29final.log`, `polar_lammps_md_mpi2_n32.log`.

Gate	Metric	Tolerance	Measured	Verdict
LAMMPS single point, 1 rank	dE vs engine	abs <= 5e-5 eV	2.910e-11 eV	PASS
LAMMPS single point, 1 rank	max dF vs engine	<= 1e-4 eV/A	2.622e-14 eV/A	PASS
LAMMPS single point, 1 rank	max position drift	monitored	4.974e-14 A	PASS
LAMMPS single point, exact 2 rank	dE vs engine	abs <= 5e-5 eV	2.910e-11 eV	PASS
LAMMPS single point, exact 2 rank	max dF vs engine	<= 1e-4 eV/A	1.687e-6 eV/A	PASS
LAMMPS single point, 1-hop 2 rank	dE vs engine	abs <= 5e-5 eV	7.937e-5 eV	FAIL
LAMMPS single point, 1-hop 2 rank	max dF vs engine	<= 1e-4 eV/A	4.116e-3 eV/A	FAIL
LAMMPS NVT MD, 128 atoms	completion / lost atoms / finite	50/50, none, finite	50/50, none, finite	PASS
LAMMPS NVT MD, 128 atoms	Etot span (relative to first step)	monitored	0.21295 eV (25 fs)	PASS
LAMMPS NVT MD, 128 atoms	temperature	300 K target	mean 276.81 K, min 188.96, max 335.95	PASS
LAMMPS NVT MD, 128 atoms	pressure	monitored	657.5 - 3231.3 bar	PASS
LAMMPS NVT MD, 128 atoms	throughput	monitored	493.662 ms/step	PASS
MPI NVT MD, 32 atoms	completion / lost atoms / Etot span	20 steps, none	20/20, none, 0.013359 eV	PASS
MPI 4 rank, 128 atoms	runnable on one GPU	must fit device memory	ghost workspace 4.5-6.9 GiB; OOM	FAIL (resource)

The 1-hop mode is 2.86x faster than exact 2-hop (1809.3 vs 5167.05 ms) but fails both the energy and force gates; it is not a usable approximation.

3. Speed

3.1 Single-rank v28 vs v29

Source: `mlip/bench/polar_perf_v28.json` and `results/polar_perf_v29.json` (20 evals each, median, energy+forces, 128-atom NH3 frame 0, 1024-atom 2x2x2 supercell).

Configuration	v28 (ms/eval)	v29 (ms/eval)	Speedup	Delta
128 atoms, deterministic	6966.79	183.78	37.91x	-6783.0 ms
128 atoms, RBE p = 50	6995.58	177.60	39.39x	-6818.0 ms
128 atoms, RBE p = 200	6962.87	186.30	37.38x	-6776.6 ms
128 atoms, RBE p = 800	7019.89	219.05	32.05x	-6800.8 ms
1024 atoms, deterministic	7795.59	1007.72	7.74x	-6787.9 ms

Model load also dropped from ~48.0 s (v28) to ~12.4 s (v29) for the deterministic case (12.95 / 12.91 / 12.89 s for RBE p=50/200/800).

Data-quality note. The stage report quotes 6951.7 ms for the v28 deterministic row. That is the *energy-only* median in `perf_breakdown_v28.log`; the benchmark JSON for the same 20-eval energy+force run is 6966.79 ms. This report uses the JSON value so all v28/v29 rows use the same measurement definition.

3.2 Same-model ablation: backbone vs full PolarMACE

Source: `results/engine_compare.json` (LAMMPS single-step `Loop time`, 128-atom NH3 box, one A100).

Configuration	Model	ms/eval	Relative to backbone
Nonlinear backbone only (polar blocks off)	polar_symmetrix_probe_polaroff.json	87.24	1.00x
Full PolarMACE v29	polar_symmetrix_probe.json	183.50	2.10x
Polar layer increment (full - backbone)	same model, polar on vs off	96.26	52.5% of full
leoff variant	polar_symmetrix_probe_leoff.json	pending	model exists (397,979,620 B); no timing JSON yet

This is the clean same-model isolation: the nonlinear backbone is 87.2 ms and the polar layer adds 96.3 ms, i.e. the full model is 2.10x the backbone. The "leoff" variant (local-electron off) has a model file but no completed timing run, so it is marked pending rather than estimated.

3.3 Linear MACE-OFF23 materialized vs generic streamed

Not the same model as PolarMACE. This table isolates the *engine path* on one linear model (MACE-OFF23 medium: 37.33 MB, 10 elements, r_cut 5 A).

Path	ms/eval	Notes
Materialized	6.6102	explicit streamed_edges materialized
Generic streamed	19.2914	requested auto, resolved generic
Auto -> generic (separate engine-compare run)	7.219	same resolution, 2.67x faster than the explicit generic run

Observed run-to-run spread on the generic path is large at this small system (7.2 vs 19.3 ms). Materialized is consistently the fastest linear path here (6.61 ms), so "streamed vs materialized" is not the explanation for the linear-vs-PolarMACE gap. The correct reading of the old 7.2 ms vs 183.5 ms comparison is "linear 37 MB/r_cut 5 A model + MH-1-capable engine path" versus "270 MB backbone + nonlinear materialized path + polar layer"; it is **not** an apples-to-apples engine or accuracy comparison. Both engines are Kokkos/CUDA.

3.4 MPI: functional, but slower on one shared GPU

Sources: `results/polar_time_mpi1.json`, `polar_time_mpi2.json`, `polar_time_mpi2_onehop.json`, `linear_mpi.json`.

Mode	128-atom eval (ms)	Relative to 1 rank	Force gate	Verdict
1 rank optimized	183.59	1.00x	PASS	production path
2 rank exact (2-hop ghost)	5167.05	0.036x (28.1x slower)	max dF 1.687e-6 eV/A, PASS	functionally correct
2 rank 1-hop ghost	1809.30	0.102x (9.85x slower)	max dF 4.116e-3 eV/A, FAIL	rejected
Upstream linear MACE H1 message passing, 1 rank	13.586	1.00x	n/a (linear model)	reference
Upstream linear MACE H1 message passing, 2 rank	45.878	3.38x slower	n/a	reference

The exact 2-hop nonlinear path is ~113x slower than the upstream linear H1 message-passing path (5167.05 vs 45.878 ms), but they are different models and different algorithms. On one shared A100, even the mature upstream linear MPI path is 3.38x slower at 2 ranks. True scaling requires a local graph with per-layer H1 feature exchange plus multiple GPUs.

3.5 Homogeneous 216-molecule per-step timings and extrapolations

Source: `mlip/bench/lammps_parity/homog_timing/homog_timing.json` (216 molecules, molecular density 0.006 A^-3, 0.5 fs timestep, 10-step LAMMPS runs, full PolarMACE v29). The N2 box is 33.019 A.

Molecule	Atoms	Temperature (K)	ms/step	10 ps	100 ps	1 ns
N2	432	151.43 (T/Tc=1.20)	476.668	2.648 h	26.482 h	11.034 d
CO2	648	364.95 (T/Tc=1.20)	1749.14	9.717 h	97.174 h	40.489 d
CH4	1080	317.00 (T*=2)	1095.37	6.085 h	60.854 h	25.356 d
H2O	648	776.52 (T/Tc=1.20)	3649.87	20.277 h	202.771 h	84.488 d
NH3	864	486.48 (T/Tc=1.20)	2217.59	12.320 h	123.199 h	51.333 d

Extrapolations assume a 0.5 fs timestep and 100% duty cycle (10 ps = 20,000 steps; 100 ps = 200,000; 1 ns = 2,000,000). They are throughput estimates, not measured long runs. An earlier pilot log (`polar_homog_timing.log`) failed the N2 case; the completed JSON above is the valid run.

3.6 RBE on the 33 A N2 box

Source: `mlip/bench/lammps_parity/rbe_md_timing/rbe_md_timing.json` (432 atoms, N2 x216, 33.019 A cubic box, 10-step LAMMPS runs, per-step time).

Mode	ms/step	Relative to deterministic
Deterministic (full k-space)	477.827	1.000x
RBE p = 50	385.427	0.807x (-19.3%)
RBE p = 200	405.901	0.849x (-15.1%)
RBE p = 800	465.281	0.974x (-2.6%)

Caveat (gate-as-minimum rule). These timings look attractive, especially p=50, but the same RBE approximation has force RMS errors of **4.099e-2 / 2.143e-2 / 1.028e-2 eV/A** at p=50/200/800 (Section 2.2), far above the **1e-4 eV/A** force gate. RBE therefore cannot be used for production force/MD work under the rule that a speed mode must pass every accuracy gate. A dedicated fixed-configuration force-error measurement on the 33 A box (`polar_rbe_force_error_33a.py`) is **in flight**; its JSON did not exist at the evidence cut-off, so no 33 A-specific force-error numbers are claimed here.

3.7 PyTorch MACE-Polar reference speed

Pending. `bench/polar_torch_reference_speed.py` exists (added 2026-10-19 08:14) but there is no `polar_torch_reference_speed.log` and no JSON output. No PyTorch-vs-engine speed ratio is claimed.

3.8 Figures

The reading package adds the higher-level diagnostic figures (relative paths from this `docs/` directory):

See reading guide for the reading order, data provenance, and reproduction entry points.

4. Findings

4.1 Cost structure: backbone plus polar non-k kernels

Completed measurements give a consistent picture:

- **128 atoms, same model:** backbone-only 87.24 ms -> full 183.50 ms. The polar layer adds 96.26 ms (52.5% of the full eval, 2.10x).
- **432 atoms, N2/33 A, same model:** full 475.883 ms/step vs polar-off 190.738 ms/step. The polar layer adds **285.145 ms (59.9%)**; the backbone accounts for 190.738 ms (40.1%).
- **Stage profile at 128 atoms** (`polar_profile_8eval_v29.log`, representative uncontended eval): backbone + stage 1 ~36.6 ms; polar forward pipeline ~20 ms, of which the k-space Coulomb stage is ~14.0 ms; reverse/replay and node forces ~53 ms. The k-space slice is therefore only ~7.6% of that eval, while the nonlinear MLP/UVU/replay kernels dominate the polar cost.

Long-range share. The full polar layer is ~60% of the 432-atom step in the completed coarse ablation. The finer figure that the ongoing profiling round attributes to the k-space/long-range slice, **~20% of the 432-atom step**, is an in-flight estimate: the dedicated 432-atom stage profile (`polar_profile_n2_33a.py`) had produced no output at the evidence cut-off. Treat the ~20% as provisional until that JSON/report exists; the 60% full-polar ablation and the 40% backbone share are directly measured.

Bottleneck: the nonlinear backbone (roughly 40-48%) plus the polar non-k kernels (UVU/MLP reverse, replay, local-electron, node forces). Optimizing only the k-space Ewald/RBE stage cannot remove most of the cost.

4.2 MH-1 is architecturally blocked

The rejection is a family-shape check, not a missing export or a flag. The model's second interaction differs on edge channels, hidden irreps, product output, and readout class; the checkpoint matches the JSON exactly. Route A (re-export) is infeasible without retraining/distillation, and Route C (relaxing only the gate) would run unprepared conditioned-MLP tables and is numerically wrong. The staged route estimates are engineering estimates, not measurements:

Route	Scope	Estimated speedup	Effort
B1	**family analysis + pair conditioning + generic streamed, FP32 enablement**	**~0.5-1.0x (not a speed guarantee)**	**5-8 days**
B2	**B1 + v4 contract + JIT host/device artifacts, polar materialized**	**target 1.5-3x backbone, ~1.2-1.5x full model**	**+15-25 days**
B3	**full polar direct/JIT**	**up to ~1.5-2x full model**	**+10-20 days**
B4	**LAMMPS direct/JIT wiring**	**required to change the measured LAMMPS timings**	**+10-15+ days**

None of these estimates is included in the measured tables above.

4.3 RBE and 1-hop are rejected by the force gate

Both are genuine performance ideas, and both fail the minimum-accuracy rule:

- RBE: excellent statistical behavior (slope -0.49885, unbiased, bitwise same-seed reproducibility) but RMS_F 1e-2 to 4e-2 eV/A, i.e. 100-400x the 1e-4 eV/A force gate.
- 1-hop ghost: 2.86x faster than exact 2-hop, but dE 7.94e-5 eV (1.6x over the 5e-5 eV tolerance) and max dF 4.116e-3 eV/A (41x over the 1e-4 eV/A gate).
- Exact 2-hop is the only MPI mode that passes, and it is 28.1x slower than 1 rank on one GPU.

4.4 One GPU means MPI does not speed up

All MPI timings above were taken on one shared A100. The exact 2-hop ghost graph has ~20x more edges than the 1-rank graph in the 14.7 A box, and the 2-hop workspace exceeds a single card at 4 ranks (4.5-6.9 GiB per block). The upstream linear H1 message passing is also 3.38x slower at 2 ranks on the same GPU. MPI is a functional/correctness milestone, not a speedup, until per-layer feature exchange and multiple GPUs are available.

5. In-flight optimization round (no results claimed)

Other agents are actively optimizing the polar and backbone kernels. The following workstreams were observed on disk at the evidence cut-off. None is included in any number above.

Workstream	Evidence	Current state	Reporting
Polar kernel optimization	**rebuild logs symmetrix_rebuild_v29alloc*.log, ..._v29lealloc.log, ..._v29tune.log; no new timing JSON**	**in flight**	**pending**
Backbone kernel optimization	**symmetrix_rebuild_v31bb1.log (08:18)**	**in flight**	**pending**
ncu / kernel profile	**bench/kernel_profile_2026-10-19/; runs/full_v29/cupti_kernels.csv has header only**	**in flight**	**pending**
MPI H1 message passing	**logs/p5_lammps_build_v31h1.log; LAMMPS built successfully at 07:49, but no parity or timing JSON**	**in flight; build success is not a result**	**pending**
432-atom stage profile	**bench/polar_profile_n2_33a.out is 0 bytes**	**in flight**	**pending**

Workstream	Evidence	Current state	Reporting
33 A RBE force-error measurement	bench/polar_rbe_force_error_33a.py (08:15), no output JSON	in flight	pending
PyTorch reference speed	bench/polar_torch_reference_speed.py (08:14), no log	in flight	pending
leoff timing	model file exists; no timing JSON	not yet run	pending

No engine source, model file, or agent bench output was modified while preparing this report.

6. Reproduction commands

All commands run from the MLIP bench workspace, which is the source of the gates and logs:

```
export MLIP=/workspace/group/shared/bza/task3_sampling_database/mlip
cd "$MLIP"

export CUDA_VISIBLE_DEVICES=0
export SYMMETRIX_BACKEND=cuda12-sm80
export SYMMETRIX_BACKEND_DESCRIPTOR_PATH="$MLIP/models/mace-polar/backend_cuda12_sm80.json"
export
LD_LIBRARY_PATH=/opt/matos-tools/cuda/12.9/lib64:/opt/matos-tools/conda/envs/matos-compute/lib
PY=/workspace/caches/conda/envs/polar/bin/python

# Accuracy gates
$PY bench/polar_module5_gate.py
$PY bench/polar_module6_gate.py
$PY bench/polar_module7_gate.py
$PY bench/polar_module8_gate.py
$PY bench/polar_endtoend_parity.py
$PY bench/polar_rbe_gate.py

# LAMMPS pair style and MD (requires the patched LAMMPS build)
$PY bench/polar_lammps_parity.py
$PY bench/polar_lammps_md.py

# Speed
$PY bench/polar_perf_report.py
$PY bench/polar_vs_linear_speed.py
$PY bench/polar_backbone_lammps.py
$PY bench/linear_streamed_compare.py
$PY bench/linear_mpi_speed.py
$PY bench/polar_homog_timing.py
$PY bench/polar_rbe_md_timing.py

# Pending / in-flight (no completed artifact at the cut-off)
$PY bench/polar_rbe_force_error_33a.py
$PY bench/polar_profile_n2_33a.py
$PY bench/polar_torch_reference_speed.py
```

Notes: - Do not run these on a shared GPU without exclusive access; the v29 absolute times are contention-sensitive. - The patched LAMMPS binary used by the gates is `/workspace/group/shared/bza/external-repos/symmetrix-xl/symmetrix/build-lammps-aacd036df381800a/lmp`. - The engine patch is `patches/symmetrix-xl-polar-support.patch` (325,145 bytes).

7. Artifact index

7.1 In this repository (links are relative to `docs/`)

Artifact	Path
v29 performance JSON	../results/polar_perf_v29.json
v28 vs v29 figure	../results/polar_perf_v28_vs_v29.png
Engine / model ablation JSON	../results/engine_compare.json
Engine comparison figure	../results/polar_engine_compare.png
Streamed vs materialized JSON	../results/streamed_compare.json

Artifact	Path
MPI 1-rank JSON	../results/polar_time_mpi1.json
MPI exact 2-hop JSON	../results/polar_time_mpi2.json
MPI 1-hop JSON	../results/polar_time_mpi2_onehop.json
Linear H1 MPI JSON	../results/linear_mpi.json
MPI status figure	../results/polar_mpi_status.png
LAMMPS parity, 1 rank	../results/polar_lammps_parity.json
LAMMPS parity, 2 rank	../results/polar_lammps_parity_mpi2.json
LAMMPS parity figure	../results/polar_lammps_parity.png
LAMMPS NVT MD JSON	../results/polar_md.json
LAMMPS NVT MD figure	../results/polar_lammps_md.png
RBE gate JSON	../results/polar_rbe_gate_v29.json
RBE gate figure	../results/polar_rbe_gate_v29.png
MH-1 feasibility analysis	design/MH1_FASTPATH_FEASIBILITY.md
Key numeric rows (CSV)	../results/comparison_summary.csv

7.2 External source artifacts (workspace paths)

Artifact	Path
v28 benchmark JSON	/workspace/group/shared/bza/task3_sampling_database/mlip/bench/polar_perf_v28.json
v29 benchmark JSON (copy)	/workspace/group/shared/bza/task3_sampling_database/mlip/bench/polar_perf.json
Engine comparison JSON	.../bench/lammps_parity/engine_compare/engine_compare.json
Homogeneous timings	.../bench/lammps_parity/homog_timing/homog_timing.json
RBE 33 A timings	.../bench/lammps_parity/rbe_md_timing/rbe_md_timing.json
Backbone ablation (N2)	.../bench/lammps_parity/backbone_timing/result.n2_full.json, result.n2_polaroff.json
Gate logs	.../mlip/logs/polar_module{5,6,7,8}_gate_v29*.log
End-to-end log	.../mlip/logs/polar_endtoend_v29final.log
RBE gate log	.../mlip/logs/polar_rbe_gate_v29opt.log
LAMMPS parity/MD logs	.../mlip/logs/polar_lammps_parity_v29final.log, polar_lammps_md_v29final.log, polar_lammps_md_mpi2_n32.log
Stage profile log	.../mlip/logs/polar_profile_8eval_v29.log
Engine patch	patches/symmetrix-xl-polar-support.patch

8. Data-quality notes and missing artifacts

These are deliberate caveats, not inferred results:

1. **`results/polar_perf_v28.json` is absent from this repository.** The v28 benchmark JSON exists only at `mlip/bench/polar_perf_v28.json`; the v28 numbers in Section 3.1 come from there.
2. **v28 deterministic row:** the report's 6951.7 ms is an energy-only median; the energy+force benchmark JSON is 6966.79 ms. Section 3.1 uses the JSON.
3. **End-to-end relative error:** the v29 final log shows 3.2e-12 to 1.1e-11 across the 32 frames. The "<=4.2e-12" line in `REPORT_2026-10-19.md` matches only the CO2 subset; the tables above use the full 32-frame range.
4. **`results/polar_md.json` vs final log:** the JSON records 599.584 ms/step (29.9792 s loop); the final v29 log records **493.662 ms/step** (24.6831 s loop) with identical thermo. The difference is GPU contention; Section 2.3 uses the final log and notes the JSON value here.
5. **Linear generic timing spread:** the engine-compare run measured 7.219 ms with `auto -> generic`; the explicit streamed compare measured 19.291 ms on the same path. Both are reported; materialized (6.6102 ms) is the stable linear reference.
6. **`n2_homog_pilot.log` is empty (0 bytes).** The completed result is `homog_timing.json`, not that pilot log.
7. **432-atom long-range share:** the ~20% k-space estimate is in flight; the completed coarse ablation gives 59.9% for the *whole polar layer*. The dedicated profile output was empty at the cut-off.
8. **33 A RBE force-error JSON, PyTorch reference speed log, leoff timing, ncu/kernel profile data, and MPI H1 validation/timing are all pending.** No numbers were invented for them.
9. **4-rank OOM** is reported from the v30 report; no separate JSON artifact was produced.

10. Absolute timings are contention-sensitive; ratios from the same run are the reliable quantity.

Prepared from completed project artifacts only. No GPU runs, no engine edits, no builds, and no writes to other agents' bench outputs.

Figure gallery

The key diagnostic figures from docs/figures/. Full captions are in the report body and in the reading guide.

Figure 1. Where the time goes: backbone vs long-range vs polar kernels.

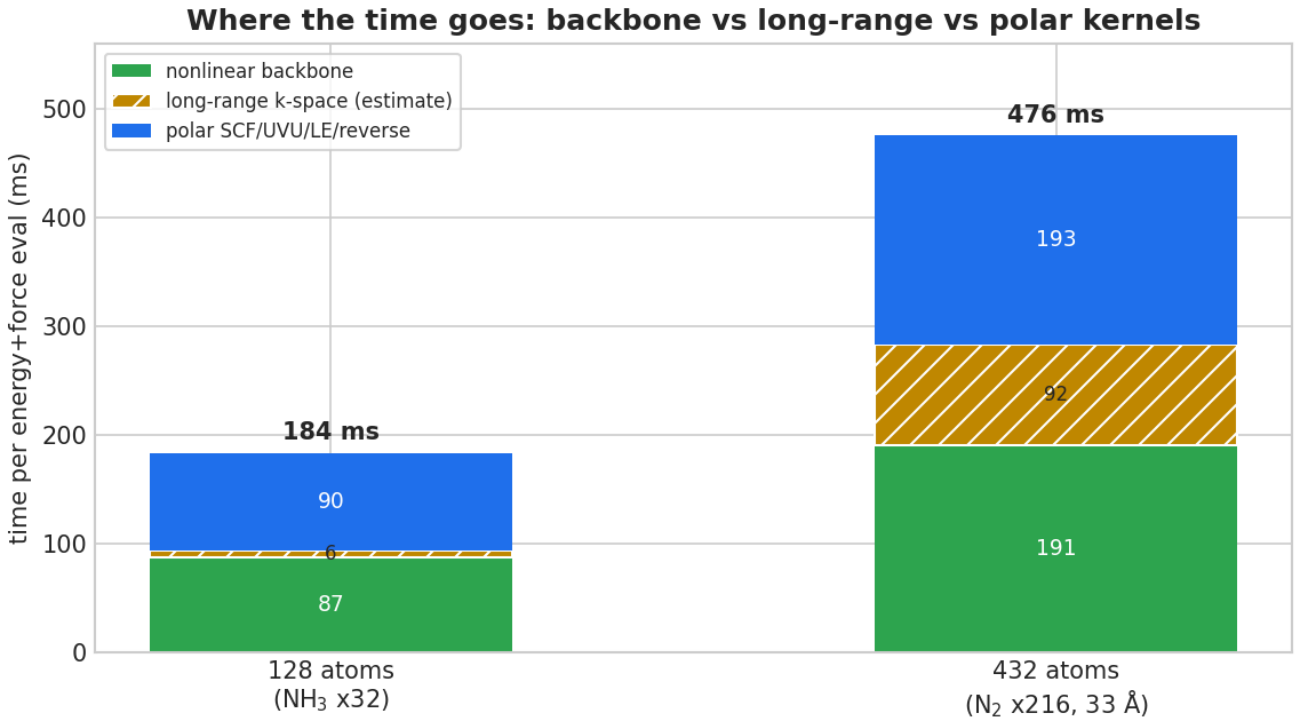


Figure 2. v28 vs v29: P5 LAMMPS wiring and the 37.8x single-rank speedup.

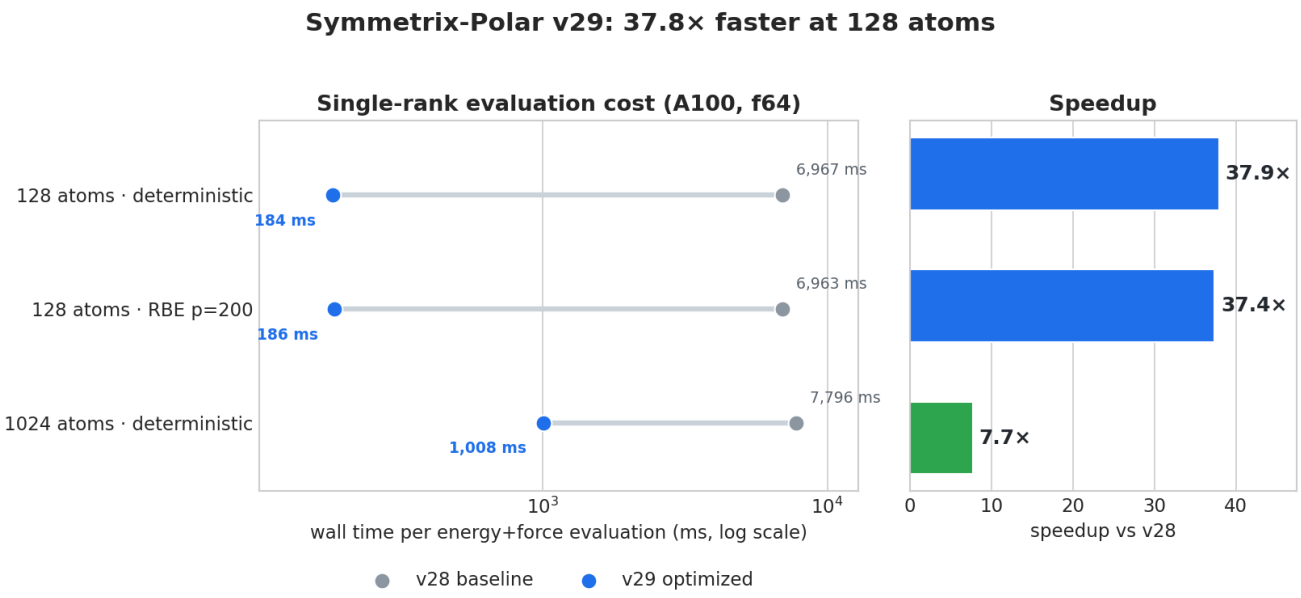


Figure 3. RBE is faster but fails the 1e-4 eV/Å force gate.

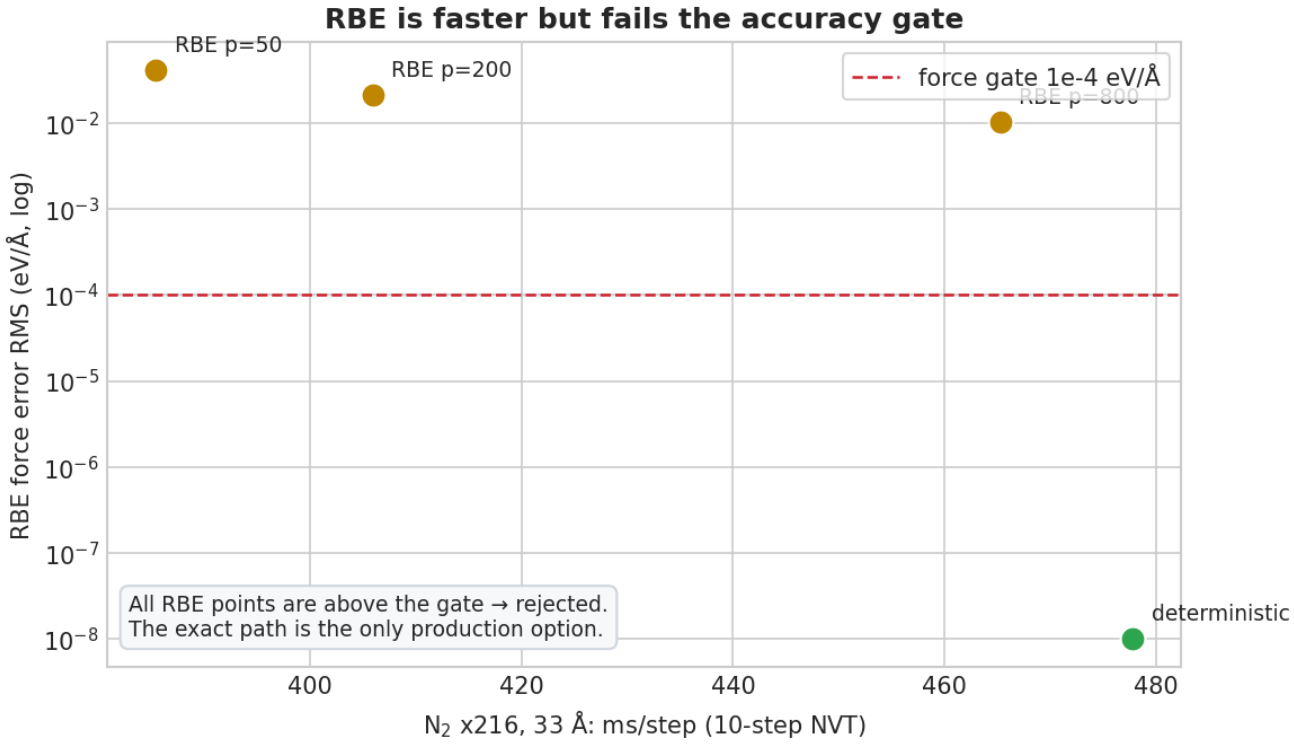


Figure 4. RBE force error vs batch size against the hard force gate.

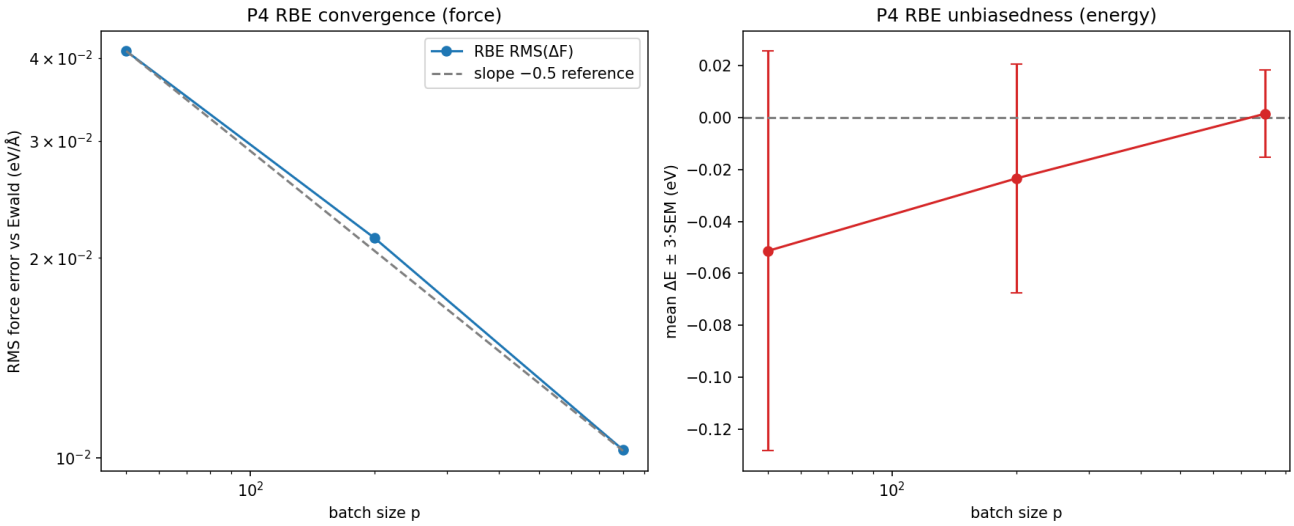


Figure 5. All accuracy gates pass with large margin.

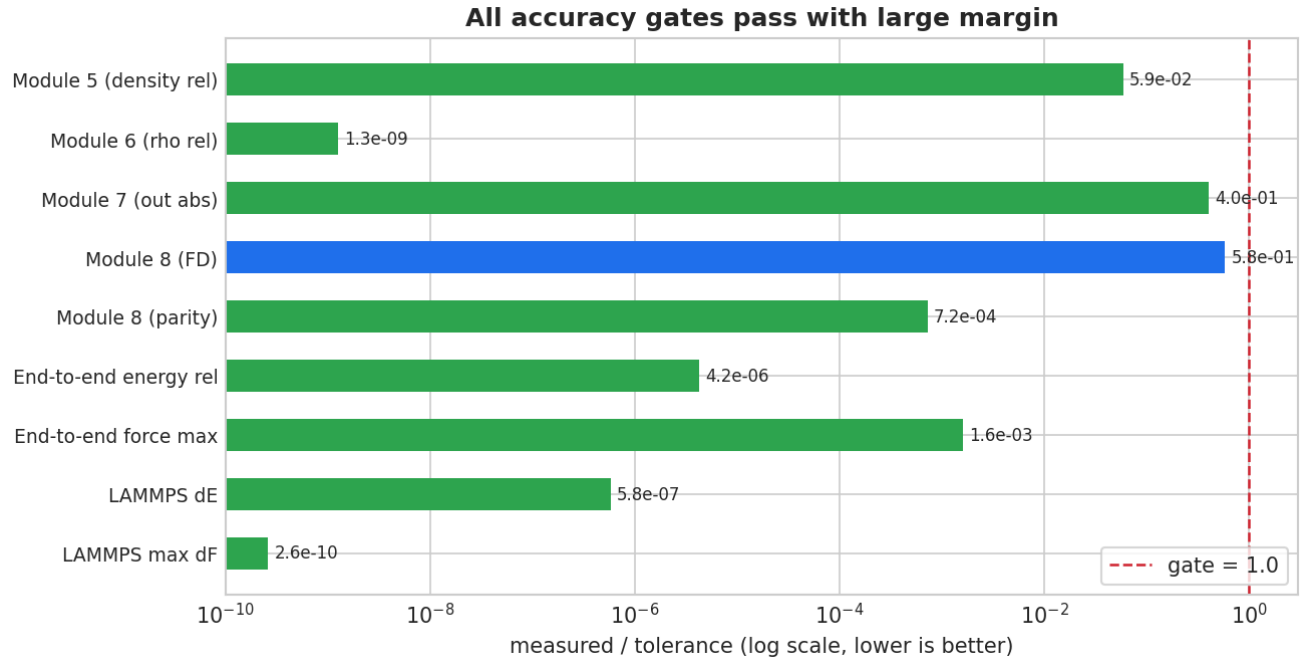


Figure 6. Homogeneous MD per-step cost and PNAS-length 1 ns extrapolation.

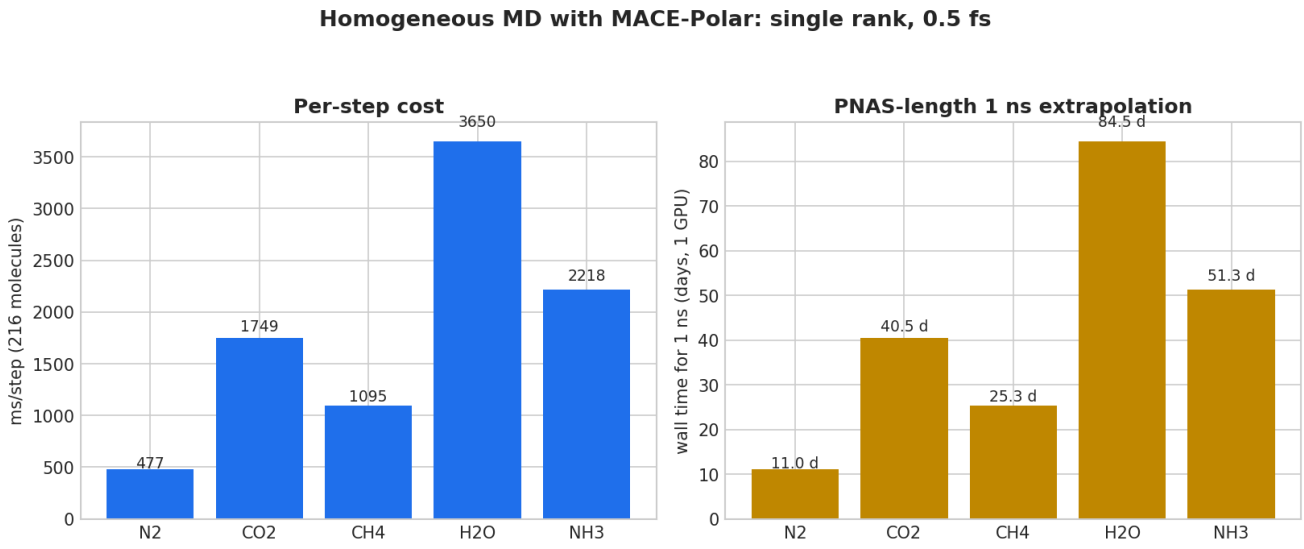


Figure 7. Engine/model cost: linear MACE vs nonlinear backbone vs full PolarMACE.

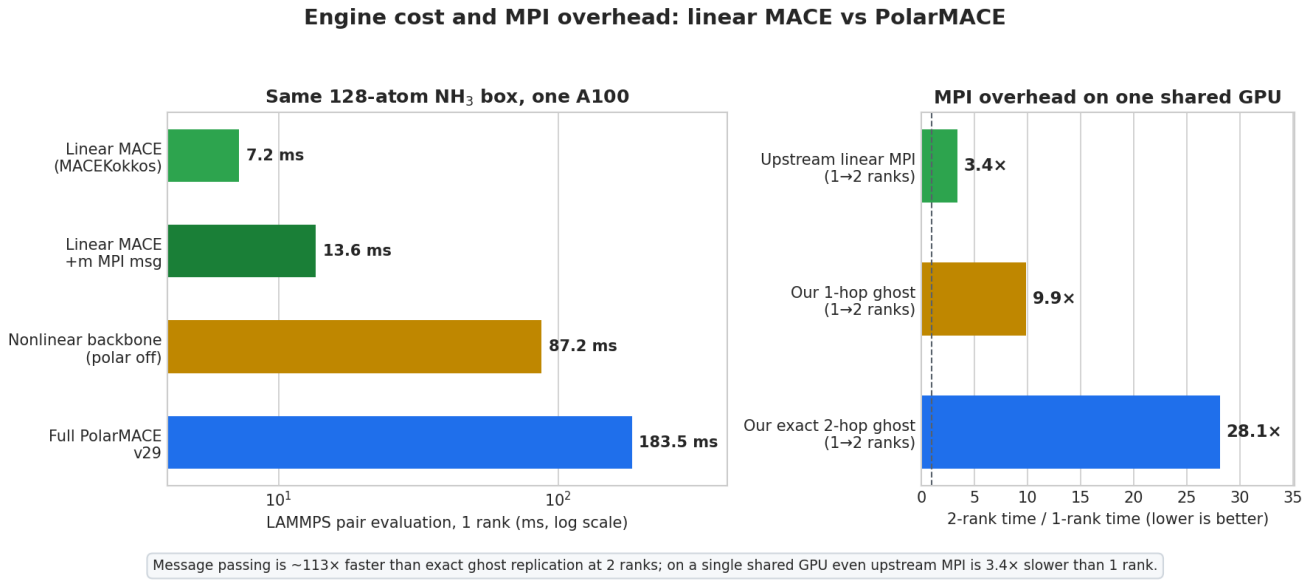


Figure 8. MPI on one shared A100: correctness achieved, no speedup yet.

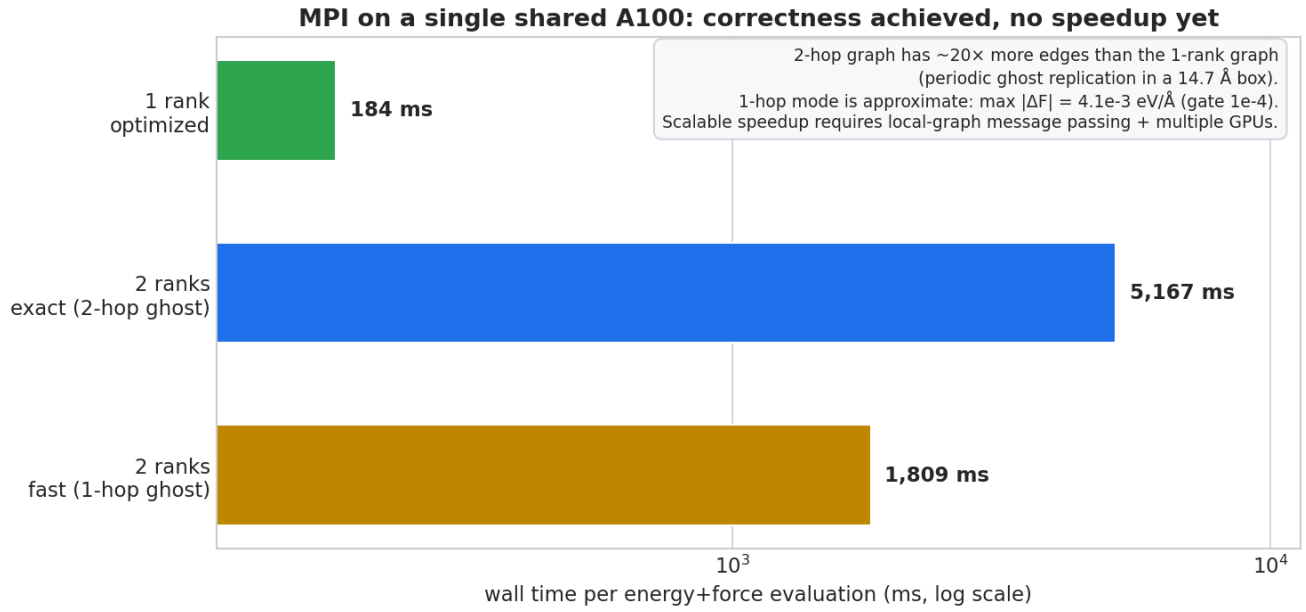


Figure 9. LAMMPS single-point force parity and residual distribution.

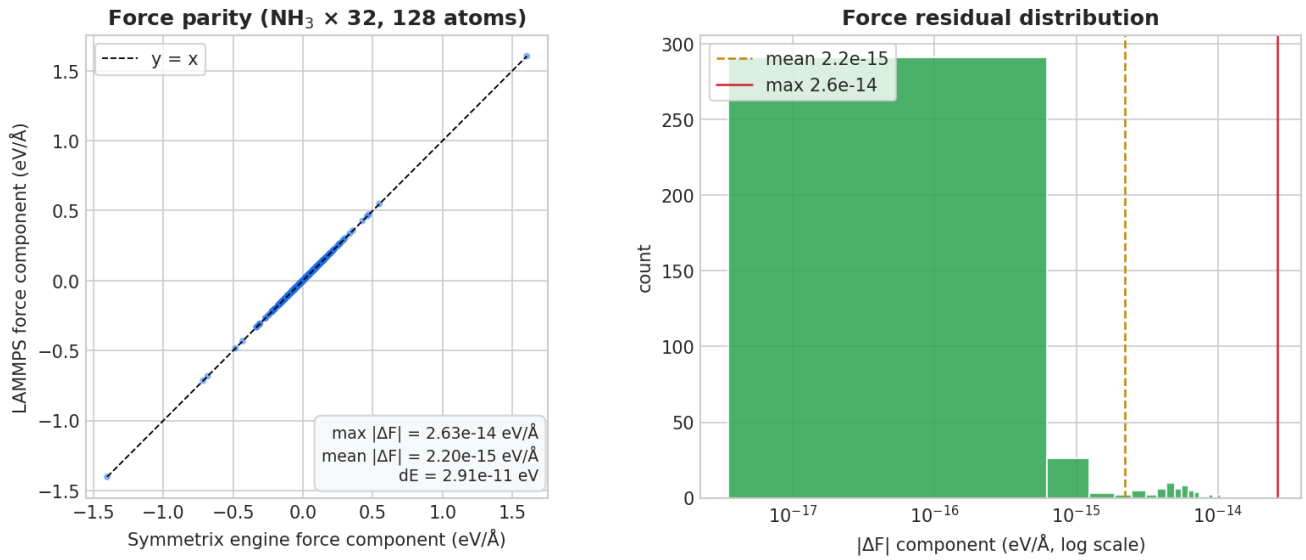


Figure 10. NVT MD temperature and energy stability.

